

GNSS ORBITS & VISIBILITY

Assigned: 4 September 2026

Due: 18 September 2026

OBJECTIVES

- Compute GPS satellite positions using an almanac and compare to precise positions from sp3
- Convert user positions between cartesian ECEF (x, y, z) and geodetic coordinates (lat, lon, height)
- Transform vectors between ECEF and local level coordinates
- Compute azimuth, elevation, and range from satellite and user positions
- Analyze satellite visibility

OVERVIEW

We will work with data for one day and several locations on the Earth:

- Date: Wednesday, August 19, 2026 = GPS Week 2432 (or 384), TOW 259200-345600/ DOY 231
- User locations:
 - NIST (in Boulder, CO): ECEF coordinates: [-1288398.567 -4721696.932 4078625.350] (m)
 - Smead Aerospace Backyard (choose reasonable coordinates)
 - EQUA (Latitude = 0, same longitude as NIST, choose a reasonable height)
- Almanac File: YUMA231.alm.txt (I added YUMA to the front of the filename)
Downloaded from <https://www.navcen.uscg.gov/archives> - on this website you choose the year & DOY
Or almanac.yuma.week0384.319488.txt
from <https://celestrak.org/GPS/almanac/Yuma/2026/> - on this website you choose the week number & TOA
- SP3 File: [IGS00PSFIN_20262310000_01D_15M_ORB.SP3](#)
Downloaded from CDDIS. *Caution – these are in km, all other positions in this HW are in meters.*
<https://cddis.nasa.gov/archive/gnss/products/2432/>

ASSIGNMENT

- 1) GPS Satellite Positions – **PRN05**
 - a. Read in the GPS satellite ECEF positions for the reference date from the sp3 file using the `read_sp3.m` function (or other means). Select the satellite specified above and plot its X, Y, Z components (in meters) vs time (GPS time shown as 0 to 24 hr into the day) as given in the sp3 file.
 - b. Now compute the GPS satellite ECEF positions for the same satellite at the same times, based on the YUMA almanac for that day. MATLAB functions for reading the file (`read_GPStyuma.m`) and computing positions (`alm2pos.m`) are provided on Canvas – you may use these or other similar utilities, or write your own. Make sure to look through the code to see how the almanac algorithm is implemented. Add these calculated values to the plots produced in part a.
 - c. In a separate figure, plot the differences between the almanac and sp3 results (almanac minus sp3) vs time.
 - d. Briefly describe the accuracy of the GPS positions predicted by the almanac. Is this accuracy sufficient to determine if a satellite should be visible to an observer on the earth? Justify your answer.

2) Ground User Position and the ENU Frame

- a. Make a table of the ECEF positions and latitude, longitude, height (wrt WGS84 ellipsoid) for all three user locations. Please express all values in meters or degrees, with 1m precision. Explain your process, check your answers, and provide well-commented computer code.

Notes: `sprintf` and `fprintf` in MATLAB are useful for specifying the number of digits to print.

To do the conversion from ECEF to LLA, you can write your own function or use the MATLAB function `ecef2lla`. Check your result by converting your result back to ECEF. You can use `ecef2lla` to do that. Check the difference between the original and final ECEF positions.

- b. Write a function that takes as input a reference latitude and longitude in degrees and returns the transformation matrix to rotate from ECEF to ENU with respect to this reference location:

`C_ECEF2ENU = ECEF2ENU(ref_lat_deg, ref_lon_deg)`

Check your result for the NIST and EQUA locations given above : First find the reference position vector in ECEF and normalize to get its unit vector $[\hat{r}]_{ECEF}$. Then use the transformation matrix you found for this location to compute $[\hat{r}]_{ENU}$. Approximately what should it to be? Explain the results for both sites.

3) Single Satellite Visibility

- a. Write a function that computes the geometrical range (m), azimuth (deg), and elevation (deg) of a GPS satellite based on the ECEF locations of a user and GPS satellite. Check your function with a simple example or compare to a reliable source. (It would be helpful if this function could handle an array of GPS satellite positions.)
`[AZ, EL, RANGE] = compute_azelrange(userECEF, satECEF)`
- b. Calculate and plot the azimuth, elevation, and range vs time for the satellite chosen in part 1a as seen by a user (observer) at NIST. (Use separate panel for each one.)
- c. Repeat the plots in part 3b showing the values only when the satellite should be visible at NIST. Briefly describe the results.
- d. Compute and plot on the same graphs the results for a second satellite that has a different pattern of visibility than the original PRN. Briefly compare and explain why they might be different.

4) Full Satellite Visibility

- a. Calculate the azimuth and elevation angles and use `plotAzEl.m` or a similar utility to show the skyplot of all GPS satellites that should be visible on the reference day from NIST.
- b. Produce similar skyplots for the other two user locations specified above.
- c. Discuss the results and how the actual environment near the antenna would impact the visibility.

5) Optional - extra ideas for those interested in exploring further (not required, worth up to +5 extra points total)

- a. For ideal circular GPS orbits and a spherical Earth, derive analytically the maximum GPS satellite elevation an observer would see at the North Pole.
- b. For ideal circular GPS orbits and a spherical Earth, derive analytically the minimum GPS satellite elevation an observer at the Equator would see looking due North (i.e. at an azimuth of 0).
- c. Use the almanac to compute visibility for more than 1 day and determine the actual time necessary for the geometry to repeat.